

PROJECT PLANNING PHASE

Electricity Consumption Analysis

Project Planning

Project planning is an essential phase that defines the activities, resources, and timeline required for the successful completion of the Electricity Consumption Analysis project.

The project was divided into multiple phases to ensure systematic development, proper documentation, timely completion, and successful deployment. The planning process included collecting the dataset, storing it in MySQL, performing SQL analysis, creating Tableau visualizations, publishing the dashboard on Tableau Public, integrating it with a Flask web application, and preparing the final documentation.

A structured weekly schedule was followed to complete all project tasks efficiently.

Project Schedule

Week	Activity
Week 1	Project topic selection and problem identification
Week 2	Dataset collection and preprocessing
Week 3	MySQL database creation and SQL query execution
Week 4	Tableau visualization development
Week 5	Dashboard and Story creation
Week 6	Flask web application development and Tableau Public publishing
Week 7	Testing, GitHub repository creation, and project validation
Week 8	Final documentation, report preparation, and project submission

Resource Planning

The following resources were utilized during the project:

Resource Type	Description
Dataset	Electricity Consumption Dataset (CSV)
Database	MySQL
SQL Tool	MySQL Workbench
Visualization Tool	Tableau Desktop
Dashboard Publishing	Tableau Public
Programming Language	Python
Web Framework	Flask
IDE	Visual Studio Code
Version Control	GitHub Repository
Documentation	Microsoft Word & PDF

Risk Management

The following potential risks were identified during the project planning phase:

- Missing or inconsistent electricity consumption data.

- SQL query execution errors.
- Difficulty in connecting Tableau with MySQL.
- Dashboard design and visualization challenges.
- Tableau Public publishing issues.
- Flask integration and deployment issues.
- Delays in report preparation and documentation.

Risk Mitigation

The identified risks were minimized through:

- Dataset validation and preprocessing.
- Regular SQL query testing.
- Verifying MySQL database connectivity.
- Continuous dashboard testing in Tableau Desktop.
- Publishing verification on Tableau Public.
- Testing Flask application locally before deployment.
- Maintaining backups of project files.
- Following a structured project timeline.

Deliverables

The project planning phase identified the following deliverables:

1. Cleaned Electricity Consumption dataset.
2. MySQL Database (electricity).
3. SQL Query File (queries.sql).
4. Tableau Workbook (Electricity_Consumption_Analysis.twb).
5. Interactive Dashboard.
6. Tableau Story with Business Insights.
7. Tableau Public Dashboard Link.
8. Flask Web Application.
9. GitHub Repository.
10. Final Project Report.
11. Performance Testing Report.

Milestones Achieved

- ✓ Dataset successfully collected and imported.
- ✓ MySQL database created.
- ✓ SQL queries executed successfully.
- ✓ Five Tableau worksheets created.
- ✓ Interactive dashboard developed.
- ✓ Tableau Story completed.
- ✓ Dashboard published on Tableau Public.
- ✓ Flask web application integrated successfully.
- ✓ GitHub repository created.
- ✓ Project documentation completed.

Expected Outcome

The planned activities ensure the successful development of an interactive Electricity Consumption Analysis system. The project provides meaningful visual insights into electricity usage through state-wise, region-wise, monthly trend, and geographical analyses. By integrating MySQL, Tableau, Tableau Public, Flask, and GitHub, the project delivers a complete end-to-end business intelligence solution that supports effective energy analysis and decision-making.